Everett Richards

ehrichards9@gmail.com · (858) 899-5834
linkedin.com/in/everett-richards · github.com/EverettRichards · everettrichards.com

Overview

I am a dedicated computer scientist who is passionate about using technology to make transportation systems more safe, efficient, and equitable. I am currently working towards my PhD in Computer Science at the University of Delaware, working in the CAR Lab with Prof. Weisong Shi. I am seeking an industry research role in applied intelligent transportation and/or robotics.

Education

University of Delaware Newark, DE
PhD Computer Science Aug 2026–Present

San Diego State University San Diego, CA
B.S. Computer Science Aug 2023–May 2026
B.S. Applied Mathematics

Most Outstanding Computer Science Graduate
Most Outstanding Mathematics & Statistics Graduate
Summa Cum Laude (4.0 GPA)

Research Experience

Doctoral Researcher – University of Delaware May–Aug 2025
Connected Autonomous Research (CAR) Lab, Prof. Weisong Shi Worcester, MA

- Working on state-of-the-art Vehicle-to-Everything (V2X) communication systems.

NSF REU Research Fellow – Worcester Polytechnic Institute May–Aug 2025
Vision, Intelligence, and System (VIS) Lab, Prof. Ziming Zhang Worcester, MA

- Conducted robustness evaluations through targeted injection of Gaussian noise into 3D LiDAR point clouds, simulating sensor degradation due to low-resolution sensors and adverse weather conditions.
- Proposed and empirically validated a novel noise-aware training curriculum, achieving up to a 40% improvement in model robustness under challenging real-world scenarios.

NSF REU Research Fellow – UC San Diego Sept 2024–June 2025
SU Computer Vision Lab, Prof. Hao Su La Jolla, CA

- Conducted experiments on the impact of Gaussian noise inject ion in robotic imitation learning models.
- Achieved statistically significant R^2 values between 0.91 and 0.99, supporting my argument that Gaussian noise induces a sigmoidal performance decay curve.

NSF REU Research Fellow – University of Delaware June–Aug 2024
Intelligent Edge Systems (IES) Lab, Prof. Lena Mashayekhy Newark, DE

- Developed and validated two algorithms to improve object detection accuracy in autonomous vehicles.
- Achieved up to 70% accuracy improvement over legacy methods, validated on a four-robot testbed.

Publications

- **E. Richards**. "Task-Aligned Stability Analysis of Vision-Language Models for Autonomous Driving Hazard Detection." 2026 International Conference on Machine Learning (ICML) Workshop on Combining Theory and Benchmarks. Seoul, South Korea, 2026.
- **E. Richards** and L. Dai. "Modeling Imitation Learning Robustness to Noisy Demonstrations via Sigmoid Degradation." 2025 IEEE MIT Undergraduate Research Technology Conference (URTC). Cambridge, MA, 2025.
- **E. Richards**, A. Lopez, J. Morales, and Z. Zhang. "From Chaos to Clarify: Strengthening 3D Collaborative Autonomous Vehicle Perception with Noise-Aware Training." 2025 IEEE MIT Undergraduate Research Technology Conference (URTC). Cambridge, MA, 2025.
- **E. Richards**, B. Thapa, and L. Mashayekhy. "Edge-Enabled Collaborative Object Detection for Real-Time Multi-Vehicle Perception." 2025 IEEE International Conference on Edge Computing and Communications (EDGE). Helsinki, Finland, 2025, pp. 13-22, doi: 10.1109/EDGE67623.2025.00011.

Instructional Experience

Teaching Assistant, Computer Science — Aug 2026–Present
University of Delaware — Newark, DE

- Teaching Assistant for the Department of Computer and Information Sciences.

Coding Camp Instructor — Jun–Aug 2026
iD Tech Camps / UC San Diego — La Jolla, CA

- Instructed students aged 10-12 on computer programming (Lua, Python) and video game development.

Tutor, Mathematics & Computer Science — Aug 2024–May 2025
SDSU Math and Science Learning Center (MSLC) — San Diego, CA

- Tutored students in computer science (Java, Python, data structures, algorithms), mathematics (calculus, discrete math, real analysis), and physics.
- Participated in professional development workshops to enhance tutoring skills.
- Certified Level 1 Tutor by the College Reading and Learning Association (CRLA).

Instructional Assistant, Discrete Mathematics — Jan 2024–May 2025
SDSU Dept. of Mathematics and Statistics – Instructor: Vadim Ponomarenko — San Diego, CA

- Led weekly office hours for ten groups of 6 to 8 students, covering proof techniques, recursion, and boolean algebra.
- Developed and discussed weekly focus exercises to reinforce key topics.

Leadership Experience

President, ACM SDSU — Apr 2024–May 2026
Association for Computing Machinery (ACM) Student Chapter at SDSU — San Diego, CA

- Founded and led SDSU's chapter of the world's largest professional organization dedicated to computing.
- Secured $2,500 sponsorship from Google to fund events and initiatives.
- Organized and personally presented dozens of technical workshops on machine learning, web development, and version control, and other topics.

Vice President, CTRL SDSU — May 2025–May 2026
Coalition of Tech Representatives and Leadership (CTRL) at SDSU — San Diego, CA

- Served on a board of leaders from various Computer Science student organizations at SDSU.
- Represented CTRL at CS faculty and Industry Advisory Board (IAB) meetings.
- Co-Directed the biannual student-led ”Innovate4SDSU” Hackathon.
- Oversaw the NXP Engineering Bootcamp, in which 15 students worked with NXP industry mentors over the course of the semester to create AI-driven solutions to chip design problems.

University Council Representative — May 2025–May 2026
Associated Students of San Diego State University — San Diego, CA

- Represent 8,000 College of Sciences students on the University Council.
- Oversee financial appropriations for a non-profit with over $45,000,000 in annual revenue.
- Drafted a successful resolution advocating for increased transparency in changes to mandatory student fees.

Workshops and Invited Talks

- **E. Richards** (2026). *Introduction to Prompt Engineering & Development with Generative AI.* Workshop hosted by the Department of Computer Science at San Diego State University.
- **E. Richards** (2025). *Imitation Learning for Robotics with Synthetic Data Generation.* Research seminar given for the NSF REU in Interdisciplinary AI, University of California San Diego.
- **E. Richards** (2025). *Python Data Science Fundamentals with NumPy and Pandas.* Workshop given for the ACM Student Chapter, San Diego State University.
- **E. Richards** (2025). *GPU Architecture with Applications in Machine Learning.* Workshop given for the ACM Student Chapter, San Diego State University.

- N. Chatlani & **E. Richards** (2024). *Undergraduate Research: Finding, Applying, Succeeding.* Workshop given for the ACM Student Chapter, San Diego State University.
- J.P. Zendejas & **E. Richards** (2024). *Introduction to Version Control with Git.* Workshop given for the ACM Student Chapter, San Diego State University.
- **E. Richards** (2024). *Introduction to Python Programming.* Workshop given for the ACM Student Chapter, San Diego State University.

Volunteering

Hackathon Judge, *CipherHacks San Diego (High School Hackathon)* — Jun 2026

Remote Competition Judge, *National Collegiate Cyber Defense Competition* — Feb 2026

Hackathon Co-Director & Project Judge, *Innovate 4 SDSU* — Apr 2025 & Nov 2025

Integration Bee Planning Committee, *SDSU Society of Physics Students* — Dec 2024

Internship Experience Panelist, *SDSU Weber Honors College* — Sept 2024 & Sept 2025

Awards and Honors

Academic Awards:

- Most Outstanding Computer Science Graduate, SDSU Dept. of Computer Science (2026)
- Most Outstanding Mathematics Graduate, SDSU Dept. of Mathematics and Statistics (2026)
- Cert. of Excellence in Advanced Programming Languages, World Computing Organization (2025)
- Dean's List, SDSU College of Sciences (2023-2026)

Scholarships:

- Steven Michael Rogers Scholarship in Mathematics (2025)
- George A. Hansen Scholarship (2025)
- Deloitte Foundation Scholarship (2024)
- Mensa Foundation Scholarship (2024)
- Raytheon Systems Scholarship (2024)
- Intuit STEM Scholarship (2023)

Skills and Strengths

Technical Skills:

- Implementing deep learning networks with PyTorch, OpenCV
- Designing parallelized algorithms that perform well across various GPU and CPU configurations
- Python programming and data analysis with NumPy, CuPy, SymPy, Pandas, MatPlotLib, and Scikit-learn
- Systems-level programming with C/C++
- Version control and collaborative coding using Git/GitHub
- Multi-modal computer vision, including CNN and ViT, using 2D (camera) and 3D (LiDAR) sensors
- Computer networking and edge computing
- Mathematical data science (compression, information theory, neural networks) and theoretical machine learning

Professional Skills:

- Research, literature review, and technical writing
- Public speaking and presentations
- Leadership and organizational management
- Collaborating with peers with diverse skill sets and backgrounds to achieve common goals
- Tutoring and mentoring students in various age groups (high school and university)
- Drafting LaTeX documents
- Publishing peer-reviewed research papers at reputable conferences (ICML, EDGE, etc.)